US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12399438
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Huang; Yu-Chih et al.

Method and apparatus for removing contamination

Abstract

Cleaning equipment for an EUV wafer chuck or clamp, which removes particles that have accumulated between burls on the surface of the wafer chuck. The equipment includes a spinning bi-polar electrode placed in proximity to the surface, which can attract and adsorb the charged particle residue therefrom using its generated symmetric electric field when the wafer chuck is not in use.

Inventors: **Huang; Yu-Chih (Hsinchu, TW), Chiou; Yu-Kai (Taipei, TW), Cheng; Chieh-Jen (Hsinchu, TW), Chen; Li-Jui (Hsinchu, TW)**

Applicant: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.** (Hsinchu, TW)

Family ID: **1000008779712**

Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD. (Hsinchu, TW)**

Appl. No.: **18/231416**

Filed: **August 08, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230375952 A1	Nov. 23, 2023

Related U.S. Application Data

division parent-doc US 17487006 20210928 US 12287589 child-doc US 18231416
us-provisional-application US 63166895 20210326

Publication Classification

Int. Cl.: **G03F7/20** (20060101); **G03F7/00** (20060101)

U.S. Cl.:

CPC **G03F7/70925** (20130101);

Field of Classification Search

CPC: G03F (7/70925); G03F (7/707); G03F (7/70933); G03F (7/2004)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4744833	12/1987	Cooper et al.	N/A	N/A
5507874	12/1995	Su et al.	N/A	N/A
7041989	12/2005	Neerhof et al.	N/A	N/A
7245357	12/2006	Van Zwet et al.	N/A	N/A
9312163	12/2015	Tanaka et al.	N/A	N/A
10459353	12/2018	Chien et al.	N/A	N/A
11016402	12/2020	Van Duivenbode et al.	N/A	N/A
2004/0012767	12/2003	Van Elp et al.	N/A	N/A
2005/0128459	12/2004	Zwet et al.	N/A	N/A
2010/0136218	12/2009	Parkhe	N/A	N/A
2011/0037960	12/2010	Scaccabarozzi et al.	N/A	N/A
2020/0058536	12/2019	Yang et al.	N/A	N/A
2020/0096881	12/2019	Van Duivenbode et al.	N/A	N/A
2020/0150549	12/2019	Stevens et al.	N/A	N/A
2020/0348606	12/2019	Perez-Falcon et al.	N/A	N/A
2020/0384509	12/2019	Azaria et al.	N/A	N/A
2022/0308465	12/2021	Huang	N/A	G03F 7/70925
2023/0375952	12/2022	Huang	N/A	G03F 7/707

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
03/098676	12/2002	WO	N/A
2015/154917	12/2014	WO	N/A
2019/042682	12/2018	WO	N/A

OTHER PUBLICATIONS

Non-Final Office Action issued in U.S. Appl. No. 17/487,006, dated Feb. 2, 2023. cited by applicant

---

Primary Examiner: Markoff; Alexander

Attorney, Agent or Firm: STUDEBAKER BRACKETT PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application of U.S. patent application Ser. No. 17/487,006 filed on Sep. 28, 2021, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/166,895 entitled "EUV WAFER TABLE CLEANING DEVICE" filed on Mar. 26, 2021, the entirety of which is hereby incorporated by reference.

BACKGROUND
(1) In semiconductor device manufacturing, various types of plasma processes are used to deposit layers of conductive and dielectric material on semiconductor substrates, and also to blanket etch and selectively etch materials from the substrate. One growing technique for semiconductor manufacturing is extreme ultraviolet (EUV) lithography, which employs scanners using light in the EUV spectrum of electromagnetic radiation, including wavelengths from about one nanometer (nm) to about 100 nm. During such processes, the substrate is affixed to a substrate chuck in a process chamber and a plasma is generated adjacent the substrate surface. Various techniques have evolved to affix the substrate to the substrate chuck. For example, an electrostatic chuck (ESC) can be used to hold the substrate during the plasma processes. The use of an electrostatic chuck eliminates the need for mechanical clamp rings, and greatly reduces the probability of forming particles by abrasion etc., which particles cause yield problems and require frequent cleaning of the apparatus.
(2) Even though the use of an electrostatic chuck reduces particle contamination, it is inevitable that small debris particles are nonetheless formed over time, and other contamination is generated within the process chamber during normal operation and/or cleaning. These particles and contamination when deposited or formed on the substrate chuck surface of an ESC increase the leakage of the heat transfer gas at the interface of the chuck surface and substrate. This leakage reduces the temperature control of the substrate and the efficiency of substrate cooling techniques. Consequently, the process chamber and the substrate chuck must be cleaned more frequently. This results in down-time for the apparatus, and requires an expensive and time consuming manual apparatus cleaning operation. Therefore, there is a need for improving the cleaning process without substantial down-time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1**A is a diagram of a lithography apparatus in accordance with some embodiments.
(3) FIG. **1**B is a diagram of a source side and a scanner side in accordance with some embodiments.
(4) FIG. **2**A is a diagram of a wafer stage in accordance with some embodiments.
(5) FIG. **2**B is a diagram of a stone cleaning process in accordance with some embodiments.
(6) FIG. **3**A is a first diagram of a spinning electrode and a wafer clamp in accordance with some embodiments.

(7) FIG. **3**B is a second diagram of a spinning electrode and a wafer clamp in accordance with some embodiments.
(8) FIG. **3**C is a diagram of alternate designs of the spinning electrode in accordance with some embodiments.
(9) FIG. **4**A is a diagram of measured contamination of the wafer clamp before stone cleaning in accordance with some embodiments.
(10) FIG. **4**B is a diagram of measured contamination of the wafer clamp after stone cleaning in accordance with some embodiments.
(11) FIG. **4**C is a diagram of measured contamination captured by the spinning electrode in accordance with some embodiments.
(12) FIG. **5**A and FIG. **5**B are diagrams of a controller in accordance with some embodiments.
(13) FIG. **6** is a flowchart of a cleaning process using the spinning electrode in accordance with some embodiments.

DETAILED DESCRIPTION

(14) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and also include embodiments in which additional features are formed between the first and second features, such that the first and second features are not in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(15) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, are used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus/device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term “made of” may mean either “comprising” or “consisting of.” In the present disclosure, a phrase “one of A, B and C” means “A, B and/or C” (A, B, C, A and B, A and C, B and C, or A, B and C), and does not mean one element from A, one element from B and one element from C, unless otherwise described.
(16) As used herein, the term “optic” is meant to be broadly construed to include, and not necessarily be limited to, one or more components which reflect and/or transmit and/or operate on incident light, and includes, but is not limited to, one or more lenses, windows, filters, wedges, prisms, grisms, gratings, transmission fibers, etalons, diffusers, homogenizers, detectors and other instrument components, apertures, axicons and mirrors including multi-layer mirrors, near-normal incidence mirrors, grazing incidence mirrors, specular reflectors, diffuse reflectors and combinations thereof. Moreover, unless otherwise specified, the term “optic,” as used herein, is not meant to be limited to components which operate solely within one or more specific wavelength range(s) such as at the EUV output light wavelength, the irradiation laser wavelength, a wavelength suitable for metrology or any other specific wavelength.
(17) In the present disclosure, the terms “mask,” “photomask,” and “reticle” are used interchangeably. In the present embodiment, the mask is a reflective mask. One embodiment of the mask includes a substrate with a suitable material, such as a low thermal expansion material or fused quartz. In various examples, the material includes TiO.sub.2 doped SiO.sub.2, or other suitable materials with low thermal expansion. The mask includes multiple reflective layers

deposited on the substrate. The multiple layers include a plurality of film pairs, such as molybdenum-silicon (Mo/Si) film pairs (e.g., a layer of molybdenum above or below a layer of silicon in each film pair). Alternatively, the multiple layers may include molybdenum-beryllium (Mo/Be) film pairs, or other suitable materials that are configurable to highly reflect the EUV light. The mask may further include a capping layer, such as ruthenium (Ru), disposed on the ML for protection. The mask further includes an absorption layer, such as a tantalum boron nitride (TaBN) layer, deposited over the multiple layers. The absorption layer is patterned to define a layer of an integrated circuit (IC). Alternatively, another reflective layer may be deposited over the multiple layers and is patterned to define a layer of an integrated circuit, thereby forming an EUV phase shift mask.

(18) In the present embodiments, the semiconductor substrate is a semiconductor wafer, such as a silicon wafer or other type of wafer to be patterned. The semiconductor substrate is coated with a resist layer sensitive to the EUV light in the present embodiment. Various components including those described above are integrated together and are operable to perform various lithography exposing processes. The lithography system may further include other modules or be integrated with (or be coupled with) other modules.

(19) A lithography system is essentially a light projection system. Light is projected through a 'mask' or 'reticle' that constitutes a blueprint of the pattern that will be printed on a workpiece. The blueprint is four times larger than the intended pattern on the wafer or chip. With the pattern encoded in the light, the system's optics shrink and focus the pattern onto a photosensitive silicon wafer. After the pattern is printed, the system moves the wafer slightly and makes another copy on the wafer. This process is repeated until the wafer is covered in patterns, completing one layer of the eventual semiconductor device. To make an entire microchip, this process will be repeated one hundred times or more, laying patterns on top of patterns. The size of the features to be printed varies depending on the layer, which means that different types of lithography systems are used for different layers, from the latest-generation EUV systems for the smallest features to older deep ultraviolet (DUV) systems for the largest.

(20) FIG. **1**A is a schematic and diagrammatic view of an EUV lithography system **10**. The EUV lithography system **10** includes an EUV radiation source apparatus **100** (sometimes referred to herein as a "source side" in reference to it or one or more of its relevant parts) to generate EUV light, an exposure tool **300**, such as a scanner, and an excitation laser source apparatus **200**. As shown in FIG. **1**A, in some embodiments, the EUV radiation source apparatus **100** and the exposure tool **300** are installed on a main floor (MF) of a clean room, while the excitation laser source apparatus **200** is installed in a base floor (BF) located under the main floor. Each of the EUV radiation source apparatus **100** and the exposure tool **300** are placed over pedestal plates PP**1** and PP**2** via dampers DP**1** and DP**2**, respectively. The EUV radiation source apparatus **100** and the exposure tool **300** are coupled to each other at a junction **330** by a coupling mechanism, which may include a focusing unit (not shown).

(21) The EUV lithography system **10** is designed to expose a resist layer to EUV light (or EUV radiation). The resist layer is a material sensitive to the EUV light. The EUV lithography system **10** employs the EUV radiation source apparatus **100** to generate EUV light having a wavelength ranging between about 1 nanometer (nm) and about 100 nm. In one embodiment, the EUV radiation source apparatus **100** generates EUV light with a wavelength centered at about 13.5 nm. In various embodiments, the EUV radiation source apparatus **100** utilizes laser produced plasma (LPP) to generate the EUV radiation.

(22) As shown in FIG. **1**A, the EUV radiation source apparatus **100** includes a target droplet generator **115** and an LPP collector **110**, enclosed by a chamber **105**. The target droplet generator **115** generates a plurality of target droplets **116**. In some embodiments, the target droplets **116** are tin (Sn) droplets. In some embodiments, the target droplets **116** have a diameter of about 30 microns (μm). In some embodiments, the target droplets **116** are generated at a rate of about fifty

droplets per second and are introduced into an excitation zone **106** at a speed of about seventy meters per second (m/s or mps). Other material can also be used for the target droplets **116**, for example, a liquid material such as a eutectic alloy containing Sn and lithium (Li).

(23) As the target droplets **116** move through the excitation zone **106**, pre-pulses (not shown) of the laser light first heat the target droplets **116** and transform them into lower-density target plumes. Then, the main pulse **232** of laser light is directed through windows or lenses (not shown) into the excitation zone **106** to transform the target plumes into an LPP. The windows or lenses are composed of a suitable material substantially transparent to the pre-pulses and the main pulse **232** of the laser. The generation of the pre-pulses and the main pulse **232** is synchronized with the generation of the target droplets **116**. In various embodiments, the pre-heat laser pulses have a spot size about 100 μm or less, and the main laser pulses have a spot size about 200-300 μm. A delay between the pre-pulse and the main pulse **232** is controlled to allow the target plume to form and to expand to an optimal size and geometry. When the main pulse **232** heats the target plume, a high-temperature LPP is generated. The LPP emits EUV radiation, which is collected by one or more mirrors of the LPP collector **110**. More particularly, the LPP collector **110** has a reflection surface that reflects and focuses the EUV radiation for the lithography exposing processes. In some embodiments, a droplet catcher **120** is installed opposite the target droplet generator **115**. The droplet catcher **120** is used for catching excess target droplets **116** for example, when one or more target droplets **116** are purposely or otherwise missed by the pre-pulses or main pulse **232**.

(24) As shown, the target droplet generator **115** generates tin droplets along a vertical axis. Each droplet is hit by a CO.sub.2 laser pre-pulse (PP). The droplet will responsively change its shape into a “pancake” during its travel along the axial direction. After a time duration (PP to MP delay time), the pancake is hit by a CO.sub.2 laser main pulse (MP) proximate to a primary focus (PF) in order to generate an EUV light pulse. The EUV light pulse is then collected by an LPP collector **100** and delivered to the exposure tool **300** for use in wafer exposure.

(25) The LPP collector **110** includes a proper coating material and shape to function as a mirror for EUV collection, reflection, and focusing. In some embodiments, the LPP collector **110** is designed to have an ellipsoidal geometry. In some embodiments, the coating material of the collector **100** is similar to the reflective multilayer of an EUV mask. In some examples, the coating material of the LPP collector **110** includes multiple layers, such as a plurality of molybdenum/silicon (Mo/Si) film pairs, and may further include a capping layer (such as ruthenium (Ru)) coated on the multiple layers to substantially reflect the EUV light.

(26) The main pulse **232** is generated by the excitation laser source apparatus **200**. In some embodiments, the excitation laser source apparatus **200** includes a pre-heat laser and a main laser. The pre-heat laser generates the pre-pulse that is used to heat or pre-heat the target droplet **116** in order to create a low-density target plume, which is subsequently heated (or reheated) by the main pulse **232**, thereby generating increased emission of EUV light.

(27) The excitation laser source apparatus **200** may include a laser generator **210**, laser guide optics **220** and a focusing apparatus **230**. In some embodiments, the laser generator **210** includes a carbon dioxide (CO.sub.2) laser source or a neodymium-doped yttrium aluminum garnet (Nd:YAG) laser source. The laser light **231** generated by the laser generator **210** is guided by the laser guide optics **220** and focused into the main pulse **232** of the excitation laser by the focusing apparatus **230**, and then introduced into the EUV radiation source apparatus **100** through one or more apertures, such as the aforementioned windows or lenses.

(28) In such an EUV radiation source apparatus **100**, the LPP generated by the main pulse **232** creates physical debris, such as ions, gases and atoms of the droplet **116**, along with the desired EUV light. In operation of the lithography system **10**, there is an accumulation of such debris on the LPP collector **110**, and such physical debris exits the chamber **105** and enters the exposure tool **300** (i.e., the “scanner side”) as well as the excitation laser source apparatus **200**.

(29) In various embodiments, a buffer gas is supplied from a first buffer gas supply **130** through the

aperture in the LPP collector **110** by which the main pulse **232** of laser light is delivered to the tin droplets **116**. In some embodiments, the buffer gas is hydrogen (H.sub.2), helium (He), argon (Ar), nitrogen (N.sub.2), or another inert gas. In certain embodiments, H.sub.2 is used, since H radicals generated by ionization of the buffer gas can also be used for cleaning purposes. Furthermore, H.sub.2 absorbs the least amount of EUV light produced by the source side, and thus absorbs the least light used by the semiconductor manufacturing operations performed in the scanner side of the lithography apparatus **10**. The buffer gas can also be provided through one or more second buffer gas supplies **135** toward the LPP collector **110** and/or around the edges of the LPP collector **110**. Further, and as described in more detail later below, the chamber **105** includes one or more gas outlets **140** so that the buffer gas is exhausted outside the chamber **105**.

(30) Hydrogen gas has low absorption of the EUV radiation. Hydrogen gas reaching to the coating surface of the LPP collector **110** reacts chemically with a metal of the target droplet **116**, thus forming a hydride, e.g., metal hydride. When Sn is used as the target droplet **116**, stannane (SnH.sub.4), which is a gaseous byproduct of the EUV generation process, is formed. The gaseous SnH.sub.4 is then pumped out through the outlet **140**. However, it is difficult to exhaust all gaseous SnH.sub.4 from the chamber and to prevent the Sn debris and SnH.sub.4 from entering the exposure tool **300** and the excitation laser source apparatus **200**. To trap the Sn, SnH.sub.4 or other debris, one or more debris collection mechanisms or devices **150** are employed in the chamber **105**. In various embodiments, a controller **500** controls the EUV lithography system **10** and/or one or more of its components shown in and described above with respect to FIG. **1**A.

(31) As shown in FIG. **1**B, the exposure tool **300** (sometimes referred to herein as the "scanner side" in reference to it or one or more of its relevant parts) includes various reflective optic components, such as convex/concave/flat mirrors, a mask holding mechanism **310** including a mask stage (i.e., a reticle stage), and wafer holding mechanism **320**. The EUV radiation generated by the EUV radiation source apparatus **100** and focused at the intermediate focus **160** is guided by the reflective optical components **305** onto a mask (not shown) secured on the reticle stage **310**, also referenced as a "mask stage" herein, within the processing chamber **306**. In various embodiments of the EUV lithography system **10**, pressure in the LPP source side is higher than pressure in the scanner side. This is because the source side uses hydrogen gas to force the removal of airborne Sn debris therefrom, while the scanner side is maintained in near vacuum in order to avoid diminishing the strength of the EUV light (being absorbed by air molecules) or otherwise interfering with the semiconductor manufacturing operations performed therein. In various embodiments, the intermediate focus **160** is disposed at the junction **330**, namely, the intersection of the source side and the scanner side.

(32) In some embodiments, the distance from the intermediate focus **160** and the reticle disposed in the scanner side is approximately 2 meters. In some embodiments, the reticle size is approximately 152 mm by 152 mm. In some embodiments, the reticle stage **310** includes an electrostatic chuck, or 'e-chuck,' to secure the mask. The EUV light patterned by the mask is used to process a wafer supported on wafer stage **320**. Because gas molecules absorb EUV light, the chambers and areas of the lithography system **10** used for EUV lithography patterning are maintained in a vacuum or a low-pressure environment to avoid EUV intensity loss. Nonetheless, a debris collector **370**, similar in purpose and design to the debris collector **150**, may be provided in the scanner side to dispose of and remove contamination and residue that may accumulate in the components of the scanner side. In various embodiments, the controller **500** controls one or more of the components of the EUV lithography system **10** as shown in and described with respect to FIG. **1**B.

(33) FIG. **2**A is a diagram of a wafer stage **320** in accordance with some embodiments. The wafer stage **320** includes a wafer clamp **322**, also referred to herein as a wafer chuck. In various embodiments, the wafer chuck **322** is an electrostatic chuck (ESC) disposed within the process chamber **306**, and the ESC is configured to receive a substrate **326**. In accordance with various embodiments, the substrate **326** includes a wafer, silicon substrate, or any other wafer, workpiece

or substrate. In various embodiments, the apparatus **10** also includes an internal chuck electrode (not shown), and a source of direct current (DC) power connected to the chuck electrode in order to provide power thereto.

(34) In various embodiments, the substrate **326** is clamped by the electrostatic chuck **322** by an electrostatic potential. In various embodiments, when a DC voltage from the source of DC power (not shown) is applied to the chuck electrode of the electrostatic chuck **322** having the substrate **326** disposed thereon, a Coulomb force is generated between the substrate **326** and the chuck electrode. The Coulomb force attracts and holds the substrate **326** on the electrostatic chuck **322** until the application of the DC voltage from the source of DC power is discontinued. In various embodiments, the applied DC voltage ranges from about 2000 volts to about 3200 volts. In some embodiments, the applied DC voltage is 3000 volts. In some embodiments, the source of DC power is configured to apply the DC voltage of about 10% to about 90% of the power applied during normal etching operations using direct current. In some embodiments, the source of DC power is configured to apply the DC voltage of about 20% to about 80%, about 30% to about 70%, or about 30% to about 50% of the power applied during normal lithography operations using direct current. In some embodiments, the application of the DC voltage occurs for a duration of about 10 seconds to about 60 seconds, about 10 seconds to about 50 seconds, about 20 seconds to about 40 seconds, or about 20 seconds to about 30 seconds.

(35) In various embodiments, the surface of the electrostatic chuck **322** includes a plurality of burls **324**, which have a width in a range from about 100 μm to about 500 μm, a height from the surface of the electrostatic chuck **322** in a range from about 1.0 μm to about 100 μm, and are spaced from each other in a range from about 1.0 μm to about 5.0 μm. In various embodiments, the electrostatic chuck **322** has greater than two thousand such burls **324**, comprising roughly 1.5% of the surface area of the surface of the electrostatic chuck **322**, in order to support the substrate **326**.

(36) In various embodiments, the application of the DC power positively charges particles or contaminants on the surface of the substrate **326**. In various embodiments, the application of the DC power negatively charges particles or contaminants on the surface of the wafer clamp **322**, which may settle upon the tops of the burls **324**. In some embodiments, debris particles or contaminants are generated from the substrate **326** during lithography operations and/or due to the electrostatic action of the wafer clamp **322**. The debris particles include, but are not limited to the following elements: Cu, Al, Ni, Ti, O, F, Si, Cu, Al, Ge, Ni, Ti, W, Xi, Mo, Fe, Pb, Bi, and In, or alloys or compounds thereof. In some embodiments, the debris particles include molecules such as: Si.sub.xO.sub.y, Al.sub.xO.sub.y, and Ti.sub.xO.sub.y. The surface debris particles and/or contaminants must be removed from the electrostatic chuck **330** in order to maximize the manufacturing performance of the apparatus **10**.

(37) FIG. **2**B is a diagram of a stone cleaning process in accordance with some embodiments. The purpose of the stone cleaning is to remove the large debris particles **340** that accumulate on top of the burls **324** of the electrostatic chuck **322** over time and with usage. In various embodiments, the stone cleaning system automatically cleans excessive focus spots caused by debris particles **340** on the burls **324** of the electrostatic chuck **322** without operator intervention. In various embodiments, the stone cleaning system uses a cleaning stone **328** in a holder (not shown) that grinds the burls **324** of the electrostatic chuck **322** as it is moved by a wafer chuck mover **380** underneath the stone **328**. In some embodiments, the stone is composed of marble. In various embodiments, the wafer clamp **322** is moved laterally (i.e., back and forth horizontally and/or vertically) and/or circularly (i.e., in expanding concentric circles or spirals) with respect to the stone **328** during this cleaning operation until the entire surface area of the wafer clamp **322** has been treated. In some embodiments, the stone cleaning process is manually performed by an operator.

(38) In various embodiments, the apparatus **10** also includes a spectral and/or charge monitoring system **385**. The spectral and/or charge monitoring system **385** is configured to monitor surface charge level and/or composition of debris particles and/or contaminants **340**, **342** on the surface of

the electrostatic chuck **322**. In some embodiments, the spectral and/or charge monitoring system **385** uses x-rays or an ion beam to charge the debris particles and/or contaminants **340**, **342** including any by-products. For example, during a stone cleaning operation of the apparatus **10**, the spectral and/or charge monitoring system **385** is configured to continuously or periodically monitor surface charge level and/or composition of debris particles and/or contaminants **340**, **342**. In some embodiments, continuous or periodic monitoring by the spectral and/or charge monitoring system **385** automatically determines when cleaning is needed and/or needs to be continued. In some embodiments, it provides a user with a contamination history or profile of the apparatus **10** throughout its service life, or any time period thereof.

(39) Through use of the spectral and/or charge monitoring system **385**, it has been discovered that the stone cleaning process can only clean the large particles **340** that exist on the top of burls **324**, and generates additional, smaller, electrically-charged particles **342** that settle between the burls **324** after stone cleaning in some embodiments. These residue particles **342** build up over time with machine usage and interfere with the efficiency and alignment of the electrostatic chuck **322**, particularly impacting wafer exposure quality, such as overlay.

(40) It has been found that the smaller residue particles **342** that settle between the burls **324** are not affected by further applications of the stone cleaning process, yet the particles **342** should nonetheless be removed. Accordingly, a spinning bi-polar electrode disk is introduced to sweep across the wafer chuck **322** (while it is inactive) in order to absorb the charged particles **342** that have settled between the burls **324** after stone cleaning in some embodiments.

(41) FIG. **3**A is a first diagram of a spinning cleaning electrode **360** provided above the wafer clamp **322** in accordance with some embodiments. In various embodiments, the wafer chuck mover **380** moves the wafer clamp **322** with respect to the spinning cleaning electrode **360**. In various embodiments, the wafer clamp **322** is moved laterally (i.e., back and forth and/or left and right horizontally and/or vertically) and/or circularly (i.e., in expanding concentric circles) with respect to the bi-polar cleaning electrode **360** while the electrode is spinning during this additional cleaning operation until the entire surface area of the wafer clamp **322** has been treated. In other embodiments, the bi-polar cleaning electrode **360** moves over the wafer clamp **322** while the wafer clamp **322** is stationary, or both the bi-polar cleaning electrode **360** and the wafer clamp **322** are moved simultaneously.

(42) The bi-polar cleaning electrode **360** is designed for electric adsorption in the near-vacuum environment of the EUV lithography apparatus **10**. In various embodiments, the bi-polar cleaning electrode **360** sweeps along the surface of the wafer clamp **322** just above the burls **324**, without physical contact. The bi-polar electrode then attracts and adsorbs both positively- and negatively-charged debris particles **342** from between the burls **342** of the wafer clamp **322** using a symmetric rotating bi-polar electric field in various embodiments.

(43) A symmetric electric field yields optimal results for particle attraction and removal of debris particles. This is because an asymmetric electric field jostles particles non-uniformly, namely in various directions including directions away from the surface of the cleaning electrode **360**. This, in turn, causes inadvertent and undesirable spreading of the debris particles from the wafer chuck surface into the processing chamber **306**.

(44) In order to generate a symmetric electric field using a shaped electrode, it is necessary to use an electrode shape that is substantially symmetrical with at least two axes of symmetry. Shapes that have a single axis of symmetry will not generate symmetric electric fields in all directions. The greater the number of axes of symmetry, the more symmetric the electric field that will be produced. A circle has infinite axes of two-dimensional symmetry. Accordingly, as shown in FIG. **3**A, the shape of the cleaning electrode **360** is circular in various embodiments.

(45) In order to generate an electric field, the surface of the cleaning electrode **360** is charged from a power source. In order to be charged, the cleaning electrode **360** is electrically conductive. In order to avoid interference with other components of the apparatus **10**, the cleaning electrode **360** is

not be magnetic or radioactive in various embodiments. Accordingly, in various embodiments, the cleaning electrode **360** is made of one or more of the following conductive elements (or molecules containing the same) in solid form: silver, copper, gold, aluminum, calcium, beryllium, rhodium, magnesium, molybdenum, tungsten, zinc, cobalt, cadmium, nickel, lithium, iron, platinum, palladium, tin, selenium, tantalum, niobium, chromium, lead, vanadium, antimony, zirconium and titanium. In some embodiments, the cleaning electrode **360** is composed of cast steel or stainless steel.

(46) Applying electric power to a conductive electrode shape will not necessarily generate a bi-polar symmetric field as needed for the present processes using a cleaning electrode. For example, simply applying opposing power leads to opposing surfaces of a circular conductive shape will yield a capacitor-like element where the top surface of the shape is, for example, positively charged, and the bottom surface of the shape is accordingly negatively charged. Such a configuration would not be optimal for cleaning electrically-charges particles **342** from the surface of the wafer chuck **322**, since the top surface would only attract negatively-charged particles while repelling the positively charged particles, and the bottom surface would only attract positively-charged particles while repelling and dislodging negatively-charged particles. In the present embodiments, both positively-charged and negatively-charged particles are distributed as debris particles **324**. Accordingly, for desirable performance in attracting both types of charged particles, the cleaning electrode **360** generates a symmetrical bi-polar electric field where a first pole is formed at one end of the surface and a second, oppositely charged pole is formed at an opposite end of the same surface from the first pole. In various embodiments, the cleaning electrode **360** has a diameter of 30±5 cm.

(47) In order to accomplish this, a first portion of the surface where the first pole resides is electrically isolated from a second portion of the surface where the second pole resides. This is accomplished in various embodiments by placing an insulator material **363** between the first and second portions. Examples of good insulating material **363** include but are not limited to: poly-vinyl chloride (PVC), glass, rigid laminates, plastic resin, polytetrafluoroethylene, an air gap and rubber. The positive lead of an electrical power supply is attached to that portion where the positive pole is desired and the negative lead of the electrical power supply is attached to the remaining portion. In various embodiments, the poles are disposed on extreme opposite ends substantially at the center of the electrode surface. Upon the application of power, such as DC power, the cleaning electrode **360** will generate a bi-polar electric field with the poles formed at the desired locations.

(48) Accordingly, with reference to FIG. **3**A, the cleaning electrode **360** has a circular shape that includes a first semicircle **362** and a second semicircle **364**, in various embodiments. In such embodiments, the first semicircle **362** is electrically insulated from the second semicircle **364** by insulating material **363**. In some embodiments, insulating material **363** is disposed entirely around a periphery of each semicircle **362**, **364**. In some embodiments, the first semicircle **362** is positively-charged by a positive lead of the power source and the second semicircle **364** is negatively-charged by a negative lead of the power source. In such embodiments, the first semicircle **362** has the positive pole and the second semicircle **364** has the negative pole of the bi-polar symmetric electric field generated by the circular shape of the cleaning electrode **360** when it is charged. In various embodiments, the positive pole and negative pole are charged to the same absolute value so as to maintain symmetry of the bi-polar electric field generated thereby. In various embodiments, the first semicircle **362** and the second semicircle **364** are equally sized and each includes approximately one-half of the shape of the cleaning electrode **360**. In other embodiments, the first semicircle **362** and second semicircle **364** are equal in size but less than one-half of the total surface area of the cleaning electrode **360**, such as ⅓, ¼, ⅛ or the like, and the remaining portion is insulating material. The charged, bi-polar symmetric cleaning electrode **360** is swept across the surface of the wafer clamp **322** in a directional manner as shown, in various embodiments, to dislodge and attract particle debris **342** from between the burls **324** thereof. The

wafer clamp mover **390** moves the wafer clamp **322** directionally (i.e., laterally and/or circularly) with respect to the cleaning electrode **360** in various embodiments. In various embodiments, the cleaning electrode **360** is placed 0.1 mm to 1 cm above the wafer clamp surface depending on the strength of the electric field (e.g., the weaker the field, the closer the cleaning electrode **360** is positioned). In various embodiments, the spectral and/or charge monitoring system **385** monitors the contaminant level of the surface of the wafer clamp **322**, and the data therefrom is used by a controller **500** or the like to automatically continue or discontinue cleaning of the wafer clamp **322**.
(49) FIG. 3B is a second diagram of the bi-polar cleaning electrode **360** in accordance with some embodiments. Simply moving a static charged symmetrical bi-polar cleaning electrode **360** will not optimally remove the particle debris **342** from the surface of the wafer clamp **322**. This is because negatively charged particles will be jostled and repelled instead of attracted by the negative semicircle **364** of the cleaning electrode **360**, and positively charged particles **342** will likewise be jostled and repelled instead of attracted by the positive semicircle **362** of the cleaning electrode **360**. Accordingly, it has been found that rotating or spinning the cleaning electrode **360** as it sweeps across the face of the wafer clamp **322** yields desirable particle removal results.
(50) In order to spin the cleaning electrode **360**, a motor **390**, such as a DC motor or an alternating current (AC) motor, is attached to the cleaning electrode **360** via an axle **391** or the like. The motor **390** may be powered from the same or a separate AC or DC power source that is used to charge the cleaning electrode **360**. The motor **390**, when powered, rotates the axle **391**, which in turn rotates the cleaning electrode **360**. The motor **390** may be rated at between one and one hundred volts, in various embodiments. In various embodiments, the motor **390** spins the electrode at between 10 and 1000 rotations per minute (RPM). In various embodiments, the cleaning electrode **360** is spun in either clockwise or counter-clockwise directions. In some embodiments, the cleaning electrode **360** may be spun in a clockwise direction for a period, followed by being spun in the opposite direction for a period during the cleaning cycle.
(51) In order to charge the cleaning electrode **360** while it is spun by the motor **390**, leads are provided through the axle **391** such that they do not get spun up and tangled during its operation. Accordingly, the leads to charge the cleaning electrode **360** may include a power transfer device **392** such as an internal hard-wired slip ring, or a wireless power device.
(52) A slip ring is an electromechanical device that allows the transmission of power and electrical signals from a stationary to a rotating structure, such as the motor **390** and the cleaning electrode **360**. The slip ring improves system operation by eliminating the need for damage-prone wires that dangle from movable joints. Also called rotary electrical interfaces, rotating electrical connectors, collectors, swivels, or electrical rotary joints. A slip ring includes a stationary graphite or metal contact (brush) which rubs on the outside diameter of a rotating metal ring. As the metal ring turns, the electric current or signal is conducted through the stationary brush to the metal ring making the connection. Additional ring/brush assemblies are stacked along the rotating axis if more than one electrical circuit is needed. Either the brushes or the rings are stationary while the other component rotates.
(53) In other embodiments, wireless power transmitters and receivers are used to provide power to charge the cleaning electrode **360** in place of slip rings. Wireless power transmitters include devices, such as inductive power transmitters that transmit power from an AC or DC power source to inductive power receivers (not shown) on or within the cleaning electrode **360**. Other methods of powering the spinning cleaning electrode **360** are readily contemplated.
(54) As shown in FIG. 3B, the spinning cleaning electrode **360** is swept over the entire surface area of the wafer clamp **322** in various embodiments. Positively and negatively charged particle **342** are attracted by the bi-polar symmetric electric field generated by the spinning cleaning electrode **360** and adsorbed (not absorbed) onto the surface of the spinning cleaning electrode **360**. The debris particles **342** are held in place on the surface of the spinning cleaning electrode **360** by coulomb forces generated by a sufficiently strong electric field in various embodiments. The process of

cleaning the surface of the wafer clamp **322** in this manner takes 1-5 complete passes of the cleaning electrode **360** over the entire surface area of the wafer clamp **322** in some embodiments. In various embodiments, the cleaning process using the cleaning electrode **360** is programmed to run for 10±2 minutes. In various embodiments, such cleaning processes are initiated periodically or after a threshold number of uses of the wafer clamp **322** during manufacturing by the apparatus **10**. The spectral and/or charge monitoring system **385** may monitor the contaminant level of the surface of the wafer clamp **322** and the data therefrom may be used by a controller or the like to automatically continue or discontinue cleaning of the wafer clamp **322** based on the residual particle debris remaining. Adsorbed debris particles **342** are shown attached to the bottom surface of the cleaning electrode **360** after being attracted off the surface of the wafer chuck **322** using the generated symmetric electric field.

(55) FIG. **3**C is a diagram of alternate designs of the symmetrical cleaning electrode **360** in accordance with some embodiments. In some embodiments, the cleaning electrode **360** is not circular, but is instead formed as a regular polygon, where all the sides of the polygon are equal, and all the interior angles are the same. Diagram **365** shows types of available regular simple polygons including squares, pentagons, hexagons, heptagons, and so forth as the number of corners of the polygon are increased. Any regular n-polygon can be used and the electric field generated thereby will be increasingly symmetrical as the numbers of corners are increased, since the number of symmetrical axes increase in proportion to the number of corners. In addition, the surface area available to adsorb debris particles **342** also increases with the increasing number of corners.

(56) In some embodiments, complex polygons can be used in place of a circular shape for the cleaning electrode **360**, since they demonstrate symmetry with two or more symmetrical axes. Complex polygons, also called self-intersecting polygons, have sides that cross over each other. The classic star is a complex polygon. Such a “regular star polygon” is a self-intersecting, equilateral equiangular polygon. A regular star polygon is denoted by its Schläfli symbol {p/q}, where p (the number of vertices) and q (the density) are relatively prime (they share no factors) and q≥2. The density of a polygon can also be called its turning number, the sum of the turn angles of all the vertices divided by 360°. Diagram **366** shows various complex polygon shapes that may be used to form the cleaning electrode **360** in various embodiments. Complex polygons will be increasingly symmetrical as the numbers of points are increased, since the number of symmetrical axes increases in proportion to the number of points. In addition, the surface area available to adsorb debris particles **342** also increases with the increasing number of points, and with increasing thickness of the same.

(57) In some embodiments, fan shapes can be used in place of a circular shape for the cleaning electrode **360**, since they also demonstrate symmetry with two or more symmetrical axes. Diagram **367** illustrates various fan shapes that may be used as the cleaning electrode **360** in place of a circular shape. Such fan shapes will be increasingly symmetrical as the numbers of “blades” are increased, since the number of symmetrical axes increase in proportion to the number of such blades. In addition, the surface area available to adsorb debris particles **342** also increases with the increasing number and thickness of the blades. In these various embodiments, the simple polygon, complex polygon and fan-shaped electrodes described above will still need to have electrically isolated first and second portions, as described with respect to circular shapes, in order to generate a symmetrical bi-polar electric field.

(58) FIG. **4**A is a diagram **402** of measured contamination of the wafer clamp before stone cleaning in accordance with some embodiments, as measured by the spectral and/or charge monitoring system **385** or the like in various embodiments. As shown therein, a large contaminant debris particle **340** has formed on an area of the surface of the wafer clamp **322** before stone cleaning is applied. In various embodiments, the circumference of such particle debris **340** is on the order of 200±50 nm.

(59) FIG. **4**B is a diagram **404** of measured contamination of the wafer clamp after stone cleaning

has been performed in accordance with some embodiments. As shown in the successive panels from left to right, the size of the debris particles **340** is decreased during application of the stone cleaning in a standard cleaning operation.

(60) FIG. **4**C is a diagram **406** of measured contamination adsorbed on the spinning cleaning electrode **360** in accordance with some embodiments, when performed after the stone cleaning. The left panel of the diagram **406** shows the smaller particle debris **342** adsorbed on the surface of the circular cleaning electrode **360**, including on the first semicircle **362** and the second semicircle **364**. The right panel shows the adsorbed particle debris **342** in greater detail. The symmetric bipolar electric field **361** generated by the cleaning electrode **360** is also shown with respect to the insulating material **363** thereon.

(61) FIG. **5**A and FIG. **5**B illustrate a computer system **500** for controlling the system **10** and its components in accordance with various embodiments of the present disclosure. FIG. **5**A is a schematic view of a computer system **500** that controls the system **10** of FIGS. **1**A-**1**B. In some embodiments, the computer system **500** is programmed to initiate a process for monitoring contamination levels of wafer holding tools using the spectral and/or charge monitoring system **385** which is in communication therewith, and provide an alert that cleaning is required. In some embodiments, manufacturing of semiconductor devices is halted in response to such an alarm. As shown in FIG. **5**A, the computer system **500** is provided with a computer **501** including an optical disk read only memory (e.g., CD-ROM or DVD-ROM) drive **505** and a magnetic disk drive **506**, a keyboard **502**, a mouse **503** (or other similar input device), and a monitor **504**.

(62) FIG. **5**B is a diagram showing an internal configuration of the computer system **500**. In FIG. **5**B, the computer **501** is provided with, in addition to the optical disk drive **505** and the magnetic disk drive **506**, one or more processors **511**, such as a micro-processor unit (MPU) or a central processing unit (CPU); a read-only memory (ROM) **512** in which a program such as a boot up program is stored; a random access memory (RAM) **513** that is connected to the processors **511** and in which a command of an application program is temporarily stored, and a temporary electronic storage area is provided; a hard disk **514** in which an application program, an operating system program, and data are stored; and a data communication bus **515** that connects the processors **511**, the ROM **512**, and the like. Note that the computer **501** may include a network card (not shown) for providing a connection to a computer network such as a local area network (LAN), wide area network (WAN) or any other useful computer network for communicating data used by the computer system **500** and the system **10**. In various embodiments, the controller **500** communicates via wireless or hardwired connection to the system **10** and its components that are described herein.

(63) The program for causing the computer system **500** to execute the process for controlling the apparatus **10** of FIGS. **1**A-**1**B, and components thereof and/or to execute the processes for manufacturing a semiconductor device according to the embodiments disclosed herein are stored in an optical disk **521** or a magnetic disk **522**, which is inserted into the optical disk drive **505** or the magnetic disk drive **506**, and transmitted to the hard disk **514**. Alternatively, the program is transmitted via a network (not shown) to the computer system **500** and stored in the hard disk **514**. At the time of execution, the program is loaded into the RAM **513**. The program is loaded from the optical disk **521** or the magnetic disk **522**, or directly from a network in various embodiments.

(64) The stored programs do not necessarily have to include, for example, an operating system (OS) or a third party program to cause the computer **501** to execute the methods disclosed herein. The program may only include a command portion to call an appropriate function (module) in a controlled mode and obtain desired results in some embodiments. In various embodiments described herein, the controller **500** is in communication with the lithography system **10** to control various functions thereof.

(65) The controller **500** is coupled to the apparatus **10** in various embodiments. The controller **500** is configured to provide control data to those system components and receive process and/or status

data from those system components. For example, the controller **500** comprises a microprocessor, a memory (e.g., volatile or non-volatile memory), and a digital I/O port capable of generating control voltages sufficient to communicate and activate inputs to the processing system, as well as monitor outputs from the lithography apparatus **10**. In addition, a program stored in the memory is utilized to control the aforementioned components of the lithography apparatus **10** according to a process recipe. Furthermore, the controller **500** is configured to analyze the process and/or status data, to compare the process and/or status data with target process and/or status data, and to use the comparison to change a process and/or control a system component. In addition, the controller **500** is configured to analyze the process and/or status data, to compare the process and/or status data with historical process and/or status data, and to use the comparison to predict, prevent, and/or declare a fault or alarm.

(66) FIG. **6** is a flowchart of a cleaning process **600** using the spinning cleaning electrode **360** in accordance with some embodiments. In some embodiments, the process **600** is controlled by the controller **500**. The cleaning process **600** commences after a lithography operation by the apparatus **10** is halted and the stone cleaning described above ends (operation **602**) in various embodiments. Next, at operation **604**, the cleaning electrode **360** is charged in various embodiments. The motor **390** is activated to spin the cleaning electrode **360** (operation **606**) in one of the clockwise or counter-clockwise directions, or in both such directions in an alternating fashion, in various embodiments. The wafer chuck **322** is then placed in proximity to the spinning cleaning electrode **360** (operation **608**) in various embodiments. In some embodiments, the wafer chuck **322** is placed under the spinning cleaning electrode **360**. In some embodiments, the wafer chuck **322** is placed under the cleaning electrode **360**, and then spinning of the cleaning electrode **360** commences. At operation **610**, the wafer chuck **322** is moved directionally under electrode across its entire surface area in various embodiments. Such movement is lateral in a horizontal and/or vertical direction, as well as circularly as described previously above in some embodiments. Next, at operation **612**, it is determined whether the wafer chuck **322** has been sufficiently cleaned. This determination is made by the controller **500** in some embodiments. The determination is made based on the duration of the cleaning process in some embodiments. The determination is made using data from the spectral and/or charge monitoring system **385** with respect to a threshold value or range in some embodiments. If the surface of the wafer chuck **322** is not sufficiently clean, the process **600** returns to operation **610** above in various embodiments. If the surface of the wafer chuck **322** is sufficiently clean, the process **600** instead continues to operation **614** where the cleaning process ends. The cleaning electrode **360** is then cleaned using debris collector **370** or the like using mechanical and/or electrical (i.e. reverse charge) processes in various embodiments. In various embodiments, the cleaning electrode **360** is cleaned in this manner without removing it from the processing chamber **306**, thereby further reducing maintenance down-time.

(67) This disclosure introduces methods and apparatus for cleaning the small electrically-charged particles **342** which accumulate between the burls **324** of a wafer chuck **322** using electrical force. Removal of such particles **342** greatly increases EUV lithographic system availability and reduces the cost of parts removal and replacement. Moreover, the manufacturing yield of the apparatus **10** is improved due to fewer overlay errors, resulting in improved exposure quality on the scanner side.

(68) According to various embodiments, a lithography apparatus includes a wafer chuck having a surface, a cleaning electrode for generating an electric field, and a motor for spinning the cleaning electrode. In such embodiments, when the cleaning electrode is spinning in proximity to the wafer chuck, electrically-charged particles are adsorbed by the cleaning electrode from the wafer chuck surface. In some embodiments, the wafer clamp is an electrostatic chuck. In some embodiments, the shape of the cleaning electrode has at least two axes of symmetry. In some embodiments, the shape is circular and the electric field is a bi-polar electric field. In some embodiments, the shape is a regular polygon and the electric field is a bi-polar electric field. In some embodiments, the shape

is one of a star shape and a fan shape, and the electric field is a bi-polar electric field. In some embodiments, the apparatus includes at least one of an alternating current (AC) power source, a direct current (DC) power source and a wireless electrical power source for charging the cleaning electrode. In some embodiments, the cleaning electrode has a first positively-charged portion and second negatively-charged portion, where the first and second portions are electrically isolated. In some embodiments, the first positively-charged portion is approximately half the shape and the second negatively-charged portion is approximately half the shape. In some embodiments, the apparatus includes an axle connecting the motor to the cleaning electrode. In some embodiments, the apparatus includes a slip ring for powering the cleaning electrode. In some embodiments, the apparatus includes a debris collector for removing the electrically-charged particles from the cleaning electrode.

(69) According to various embodiments, a method for removing contamination from a lithography device includes disposing a bi-polar electrode above a surface of a wafer clamp, charging the bi-polar electrode; spinning the bi-polar electrode; and moving at least one of the wafer clamp and the bi-polar electrode relative to each other such that electrically-charged particles on the surface of the wafer clamp are attracted onto a surface of the bi-polar electrode while the bi-polar electrode is spinning. In some embodiments, the bi-polar electrode generates a symmetric electric field. In some embodiments, the bi-polar electrode spins in one or both of the clockwise and a counterclockwise directions. In some embodiments, the wafer clamp is moved laterally and/or circularly relative to the bi-polar electrode. In some embodiments, the wafer surface is cleaned with a cleaning stone which generates the electrically-charged particles that are later cleaned by the cleaning electrode. In some embodiments, the electrically-charged particles are subsequently removed from the bi-polar electrode using a debris collector disposed within the lithography apparatus.

(70) According to various embodiments, a method for removing contamination from a lithography device includes removing large contaminant particles having a first size from a wafer chuck surface using a cleaning stone. In such embodiments, a cleaning electrode having a symmetrical surface with at least two axes of symmetry is then placed above the wafer chuck surface. In such embodiments, the cleaning electrode is then charged to generate a symmetrical electric field. In such embodiments, the cleaning electrode spins above the wafer chuck surface so that electrically-charged residue particles having a second size are adsorbed by the symmetrical surface of the electrode from the wafer chuck surface using the symmetrical electric field, wherein the first size is larger than the second size. In some embodiments, the wafer chuck surface is moved laterally and/or circularly relative to the cleaning electrode while the cleaning electrode is spinning.

(71) The foregoing outlines features of several embodiments or examples so that those skilled in the art better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A lithography apparatus comprising: a wafer chuck having a surface; a cleaning stone configured to remove large contaminant particles from the surface of the wafer chuck and generate small residue particles; a charge-monitor system configured to charge the small residue particles to generate electrically-charged small residue particles using x-rays or an ion beam; a cleaning electrode configured to generate an electric field; and a motor for spinning the cleaning electrode,

wherein when the cleaning electrode is spinning in proximity to the wafer chuck, the electrically-charged small residue particles are adsorbed by the cleaning electrode from the surface.
2. The lithography apparatus of claim 1, wherein the wafer chuck is an electrostatic chuck.
3. The lithography apparatus of claim 1, wherein the cleaning electrode comprises a shape with at least two axes of symmetry.
4. The lithography apparatus of claim 3, wherein the shape is circular and the electric field is a bi-polar electric field.
5. The lithography apparatus of claim 3, wherein the shape is a regular polygon and the electric field is a bi-polar electric field.
6. The lithography apparatus of claim 3, wherein the shape is one of a star shape and a fan shape, and wherein the electric field is a bi-polar electric field.
7. The lithography apparatus of claim 3 further comprising at least one of: an alternating current (AC) power source, a direct current (DC) power source and a wireless electrical power source for charging the cleaning electrode.
8. The lithography apparatus of claim 7, the cleaning electrode further having a first positively-charged portion and second negatively-charged portion that are electrically isolated.
9. The lithography apparatus of claim 8, wherein the first positively-charged portion comprises substantially half the shape and the second negatively-charged portion comprises substantially half the shape.
10. The lithography apparatus of claim 1, further comprising an axle connecting the motor to the cleaning electrode.
11. The lithography apparatus of claim 1, further comprising a slip ring for powering the cleaning electrode.
12. The lithography apparatus of claim 1, further comprising a debris collector for removing the electrically-charged small residue particles from the cleaning electrode.
13. A method for removing contamination from a lithography device, comprising: removing large contaminant particles from a surface of a wafer clamp by a cleaning stone and generating small residue particles; charging the small residue particles to generate electrically-charged small residue particles using x-rays or an ion beam; disposing a bi-polar electrode above the surface of the wafer clamp; charging the bi-polar electrode; spinning the bi-polar electrode; and moving at least one of the wafer clamp and the bi-polar electrode relative to each other such that electrically-charged small residue particles on the surface of the wafer clamp are attracted onto a surface of the bi-polar electrode while the bi-polar electrode is spinning.
14. The method of claim 13, further comprising: generating a symmetric electric field with the bi-polar electrode.
15. The method of claim 14, further comprising: spinning the bi-polar electrode in at least one of a clockwise and a counterclockwise direction.
16. The method of claim 14, further comprising: moving the wafer clamp at least one of laterally and circularly relative to the bi-polar electrode.
17. The method of claim 14, wherein: the removing the large contaminant particles from the surface of the wafer clamp by the cleaning stone and the generating the electrically-charged small residue particles are performed prior to said disposing the bi-polar electrode above the surface of the wafer clamp.
18. The method of claim 14, further comprising removing the electrically-charged small residue particles from the bi-polar electrode using a debris collector.
19. A lithography apparatus for cleaning a wafer chuck, comprising: a cleaning stone configured to remove large contaminant particles from a surface of the wafer chuck and generate small residue particles; a charge-monitor system configured to charge the small residue particles to generate electrically-charged small residue particles using x-rays or an ion beam; a bi-polar electrode having a symmetrical surface with at least two axes of symmetry, and comprising a first and a second pole

portions that are respectively formed at a first and a second ends of the symmetrical surface and oppositely charged; and a motor configured to spin the bi-polar electrode so that the electrically-charged small residue particles are absorbed from a surface of the wafer chuck when the bi-polar electrode is in proximity to the wafer chuck.

20. The lithography apparatus of claim 19, wherein the first and the second pole portions are isolated from each other by an insulating material on the bi-polar electrode, and wherein the insulating material comprises one of glass, plastic resin, an air gap, or rubber.